

SPECIFICATION OF ACOUSTIC TESTING AND INSPECTION FOR INSTALLED ROOMS

PART 1 GENERAL

A. SUMMARY

1. Conduct the following field test to fulfill project acoustic requirement and document submission:
 - a) Sound separation test
 - b) Background noise test
 - c) Reverberation test
2. The scope of area shall cover minimum the following area but not limited to:
 - a) Conference Rooms
 - b) Office type premises: Security Control Room & Management Offices / Network Operation Centre /Office
 - c) Other areas specified

B. DEFINITIONS

1. NIC: Noise Isolation Class
2. STC: Sound Transmission Class
3. NC: Noise Criterion
4. RT: Reverberation Time
5. Quality-Control Services: Tests, inspections, procedures, and related actions during and after execution of the Work to evaluate that actual products incorporated into the Work and completed construction comply with requirements. Services do not include contract enforcement activities performed by the Project Manager.
6. Field Quality-Control Testing: Tests and inspections that are performed on-site for installation of the Work and for completed Work.
7. Testing Agency: An entity engaged to perform specific tests, inspections, or both. Testing laboratory shall mean the same as testing agency.
8. Experienced: When used with an entity, "experienced" means having successfully completed a minimum of five previous projects similar in size and scope to this Project; being familiar with special requirements indicated; and having complied with requirements of authorities having jurisdiction.

C. REFERENCES

1. Test shall be conducted to comply the following standard:

- a) ASTM E 336 – “Standard Test Method for Measurement of Airborne Sound Insulation between Rooms in Buildings”
- b) ISO 140-4 – “Field measurements of airborne sound insulation between rooms”
- c) ASTM E 413 – “Classification for rating Sound Insulation”
- d) BS EN 15726 – “Ventilation for buildings. Air diffusion. Measurements in the occupied zone of air-conditioned/ventilated rooms to evaluate thermal and acoustic conditions”
- e) BS EN ISO 3382-2 – “Acoustics – Measurement of room acoustic parameters – Part 2: Reverberation time in ordinary rooms”

D. CONFLICTING REQUIREMENTS

1. General: If compliance with two or more standards is specified and the standards establish different or conflicting requirements for minimum quantities or quality levels, comply with the most stringent requirement. Refer uncertainties and requirements that are different, but apparently equal, to the Project Manager for a decision before proceeding.
2. Minimum Quality Levels: The quality level shown or specified shall be the minimum provided or performed. The actual installation may comply exactly with the minimum quantity or quality specified, or it may exceed the minimum within reasonable limits. To comply with these requirements, indicated numeric values are minimum or maximum, as appropriate, for the context of requirements. Refer uncertainties to Project Manager for a decision before proceeding.

E. SUBMITTALS

1. Qualification Data: For testing agencies, to demonstrate their capabilities and experience. Include proof of qualifications in the form of a recent report on the inspection of the testing agency by a recognized authority.
2. Method Statements for Acoustic Testing: Submit to the Project Manager minimum 14 days prior to the test for acceptance and include at minimum the following:
 - a) Credit numbers and titles governing work to be tested.
 - b) Description of tests.
 - c) Identification of applicable standards.
 - d) Identification of test methods.
 - e) Number of tests and inspections required.
 - f) Time schedule and time span for tests and inspections.
 - g) Entity responsible for performing tests and inspections.

- h) Unique characteristics of each test.
- 3. Reports for Responsible Field Testing: Prepare and submit certified written reports to advise if comply with project requirements and the guidelines of HKBEAM Plus and LEED that include the following. The measurement report shall be endorsed by a Corporate Member of Hong Kong Institute of Acoustics (HKIOA) or equivalent.
 - a) Date of issue.
 - b) Project title and number.
 - c) Name, address, and telephone number of testing agency.
 - d) Dates, times and locations of tests.
 - e) Equipment list with model name and serial number.
 - f) Names of individuals making tests and inspections.
 - g) Description of the work and test method.
 - h) Identification of products.
 - i) Complete test data and chart.
 - j) Test results and an interpretation of test results.
 - k) Comments or professional opinion on whether tested work complies with the Contract Document requirements, and if non-compliant observed causes and particulars of non-compliance.
 - l) Photo record for each sampling points
 - m) Certificates of calibration for test equipment, including sound level meter and acoustic calibrator.
 - n) Name and signature of laboratory inspector.
 - o) Recommendations on retesting.

F. QUALITY ASSURANCE

- 1. Testing Agency Qualifications: A HOKLAS-accredited independent agency with the experience and capability to conduct testing and inspecting indicated, with minimum 5-year equivalent acoustical measurement experience.
 - a) HOKLAS-accredited: A testing laboratory accredited under the Hong Kong Laboratory Accreditation Scheme.
 - b) The noise measurements shall be conducted by qualified acoustician(s) with minimum five years of experience in similar noise tests. The testing acoustician(s) shall be a corporate member of HKIOA, IOA, ASA or equivalent.

G. FIELD TESTING BY CONTRACTOR

1. Contractor's Responsibility:
 - a) Engage a qualified testing agency to perform specified tests. Do not employ same entity engaged by Employer, unless agreed to in writing by Employer.
 - b) Notify testing agency at least 7 days in advance of time when work that requires testing shall be ready for testing.
 - c) Testing requested by the Contractor that is not required by the Contract Documents is the Contractor's responsibility.
 - d) Costs for retesting construction that replaces or is necessitated by work that failed to comply with the Contract Documents shall be borne by the Contractor at no cost to Employer.
 - e) Check the installation work and ensure it is suitable for test. Joints shall be properly sealed and no visible gap.
2. Retesting / Reinspecting: Provide retesting for construction that replaced Work that failed to comply with the Contract Documents.
3. Testing Agency's Responsibility: Cooperate with the Project Manager and the Contractor in performance of duties. Provide qualified personnel to perform required tests and inspections.
 - a) Notify Project Manager, and the Contractor promptly of irregularities or deficiencies observed in the work during performance of its services.
 - b) The location from which test samples shall be taken and in which in-situ tests are conducted are described in section 2 below.
 - c) Conduct and interpret tests and inspections and state in each report whether tested and inspected work complies with or deviates from requirements.
 - d) Submit a certified written report, in duplicate, of each test, inspection, and similar quality-control service through the Contractor.

H. SOUND SEPARATION FIELD TEST

1. Conduct sound separation field test to determine that the tested elements have achieved specified NIC rating and the guidelines of HKBEAM Plus and LEED, i.e. separation of wall and door. Provide necessary support and coordination to facilitate sound separation field test.
2. Number of test samples: This shall be determined and agreed with the Project Manager and project acoustic consultant.
3. Commence final separation field test after the following conditions have been achieved:

- a) Suspension of ongoing construction activities which may affect acoustic data collection.
 - b) Photo records of the initial site inspection shall be carried out by the Contractor to ensure the installation was done properly.
 - c) The NIC-rated elements to be tested have been installed properly and to the standard and satisfaction of the manufacturer. Letter of acceptance on the installation of the elements shall be obtained from the manufacturer before the field test.
4. Notify the Project Manager minimum 3 days prior to attend sound separation field test witness.

I. BACKGROUND NOISE FIELD TEST

- 1. Conduct background noise field test to confirm whether the specified NC (Noise Criterion) and the guidelines of HKBEAM Plus and LEED are achieved during operation of HVAC system. Provide necessary support and coordination to facilitate background noise field test.
- 2. Number of test samples: This shall be determined and agreed with the Project Manager and project acoustic consultant.
- 3. Coordinate background noise field test with respect to construction program.
- 4. Ensure that the following conditions have been achieved before commence background noise field test:
 - a) Suspension of ongoing construction activities which may affect acoustic data collection.
 - b) Proper balancing of the heating, ventilation and air-conditioning system, including air flow, volume control damper.
 - c) Velocity and air temperature measurement have been completed and accepted by the Project Manager.

J. REVERBERATION TIME FIELD TEST

- 1. Conduct Reverberation time field test to determine that the concerned areas have achieved specified RT performance and the guidelines of HKBEAM Plus and LEED as per the installed interior finishes. Provide necessary support and coordination to facilitate the field test.
- 2. Number of test samples: This shall be determined and agreed with the Project Manager and project acoustic consultant.
- 3. A sweep signal shall be generated from the Dirac software through the source speaker and shall be picked-up by the measuring microphone at the seating position of approx. 1.2 to 1.5 m from floor in accordance to ISO 3382-2.

4. Acoustic test shall be conducted in accordance to standard acoustic practice. The measurement location will be taken at about 1.2 m above the floor. Test shall include minimum 6 points for each specified room.
5. Commence Reverberation time test after the following conditions have been achieved:
 - a) Suspension of ongoing construction activities which may affect acoustic data collection.
 - b) All the interior finishes, including audience seats and furniture have been installed.

PART 2 EXECUTION

A. ACOUSTIC FIELD TESTING

1. Compliance: the test standard as stated in Part 1 - Section C with slight adjustment to suit site condition.
2. The tested elements shall include but not limited to the following. The Project Manager has the right to instruct the Contractor for additional testing should many of the specimens fail to meet the acoustic requirement. The cost of additional testing will be borne by the Contractor.
 - a) Acoustical partitions and doors: The partitions and doors for the following rooms shall be tested to ensure the field ratings:
 - All Office type premises: Security Control Room & Management Offices / Network Operation Centre / Office
 - All Conference Rooms
 - b) Background noise test: To carry out testing on NC levels of the rooms/areas with NC 40 or below.
 - All Office type premises: Security Control Room & Management Offices / Network Operation Centre / Office
 - All Conference Rooms
 - Office Services Corridor
 - c) Reverberation time: To carry out testing on the specified rooms.
 - All Office type premises: Security Control Room & Management Offices / Network Operation Centre / Office
 - All Conference Rooms

B. TEST AND INSPECTION LOG

1. The party assigned with the responsibility of conducting acoustic field testing shall be responsible for preparing record of tests and inspections respectively, including the following:

- a) Date of test or inspection was conducted.
 - b) Description of the work tested or inspected.
 - c) Date when test or inspection results were transmitted to the Project Manager.
 - d) Identification of testing agency or special inspector conducting test or inspection.
2. Maintain log at Project Site. Post changes and modifications as they occur. Provide access to test and inspection log for Project Manager's reference during normal working hours.

C. REPAIR AND PROTECTION

1. General: On completion of testing, inspecting and similar services, repair damaged construction and restore substrates and finishes affected by testing.
- a) Provide materials and comply with installation requirements specified in other Specification sections. Restore patched areas and extend restoration into adjoining areas with durable seams that are as invisible as possible.
2. Protect construction exposed by or for testing and inspection service activities.
3. Be responsible for repair and protection, regardless of the assignment of responsibility for testing and inspection services.

END OF SECTION